

AI-Powered Mock Interviewer System

Problem Statement

In today's competitive hiring environment, candidates often struggle to prepare effectively for technical and behavioral interviews due to the lack of personalized feedback, realistic interview simulations, and affordable mentoring resources. Traditional mock interviews require human interviewers, making them time-consuming, expensive, and difficult to scale.

This project aims to develop an AI-powered Mock Interviewer System that can simulate a real interview experience by generating personalized interview questions based on a candidate's resume, evaluating the candidate's verbal responses, and analyzing behavioral indicators such as attention level and body language during the interview session.

The system will allow candidates to upload their resumes, after which the platform will automatically extract relevant skills, projects, technologies, and experience details to generate domain-specific and role-based interview questions. During the interview, the candidate's responses will be transcribed and evaluated using natural language processing techniques to assess technical accuracy, clarity, confidence, and communication skills.

Additionally, the system will use computer vision techniques through webcam input to monitor engagement metrics such as eye contact, head movement, posture, and attentiveness. Based on both verbal and behavioral analysis, the platform will generate a comprehensive performance report with scores, feedback, strengths, and areas for improvement.

The objective of this project is to create an intelligent, scalable, and interactive interview preparation platform that enhances candidate readiness through automated personalized feedback and realistic interview simulations.

Key Features

- Resume-based dynamic question generation
- AI-driven technical answer evaluation
- Speech-to-text transcription
- Attention and body language analysis
- Performance scoring and feedback generation
- Role-specific mock interview simulation
- Interview history and analytics dashboard

Expected Outcome

The final system should be capable of conducting an automated mock interview session, evaluating candidate responses and behavioral engagement, and generating detailed analytical feedback to help users improve their interview performance and confidence.